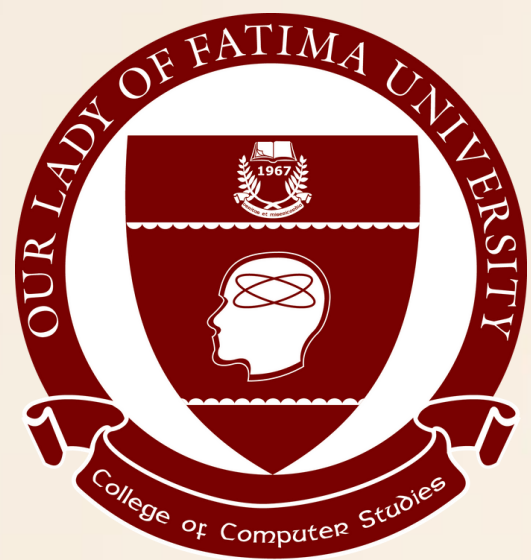




Module 9

Introduction to Ecommerce

ITCL112 - INTRODUCTION TO COMPUTING W/ LAB

INTRODUCTION



The rise of e-commerce has been driven by several factors, including the widespread availability of the internet, the proliferation of smartphones, and the development of secure online payment systems. From small startups to multinational corporations, businesses are increasingly adopting e-commerce to reach a global audience, reduce operational costs, and provide a convenient shopping experience for customers.

MODULE 9

In this module, you will learn:

- 1. » The concept and definition of Electronic Commerce**
- 2. To enumerate and differentiate the online payment systems and models**
- 3. To understand and apply security measures in E-commerce**

What is Ecommerce?

It is commonly known as Electronic Marketing. It consists of buying and selling goods and services over an electronic systems such as the internet.

Ecommerce is the purchasing, selling and exchanging goods and services over computer network or internet through which transactions or terms of sale are performed electronically



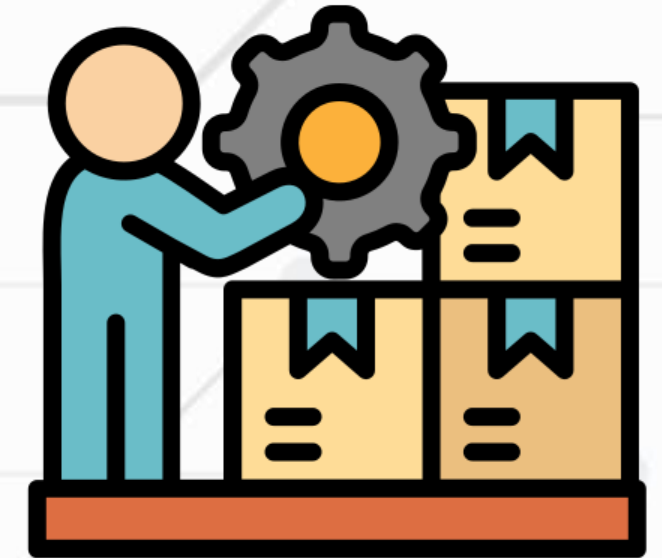
What is Ecommerce?



Buyers
- The one who has money to purchase the good



Sellers
- The one who offers good and services to buyers



Producers
- The one who creates the products and services that sellers offer to buyers

Why use Ecommerce?



- ✓ **Low entry cost**
- ✓ **Reduces transaction cost**
- ✓ **Access to the global market**
- ✓ **Secure Market share**

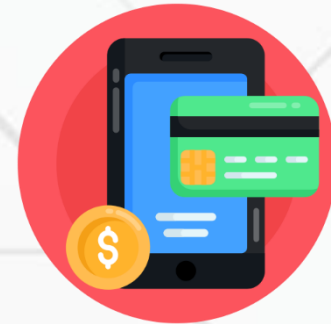
The Process of Ecommerce



E-Commerce Transaction



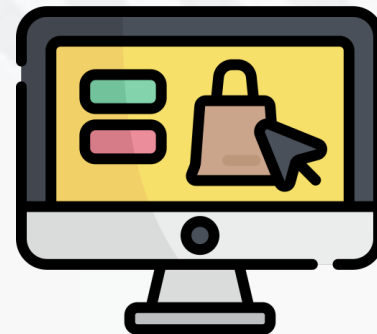
Features of Ecommerce



Non-Cash Payment – enables use of debit/credit cards, smart cards, bills payment and e-payments.



24/7 Service Availability – services are available anytime



Advertising – increases the reach of advertising of products and services of businesses



Marketing – it helps in better marketing management of products and services

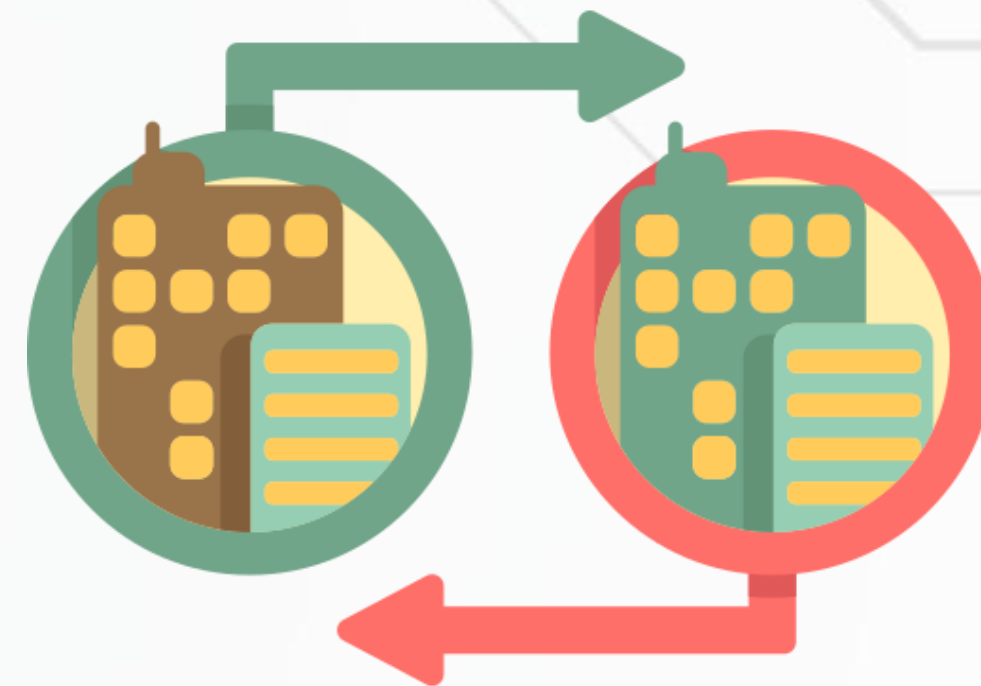
Type of Ecommerce



Business-to-Business (B2B) – is simply defined as e-commerce between companies. About 80% of e-commerce is of this type.

Examples:

- ✓ Intel selling microprocessor to Dell
- ✓ Heinz selling ketchup to McDonalds



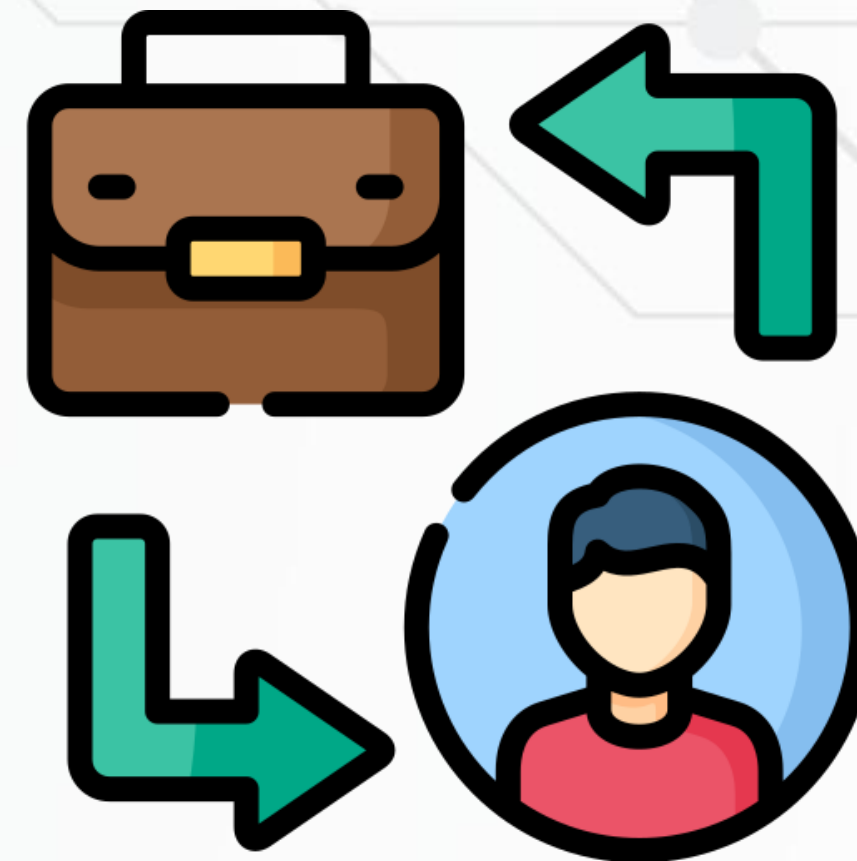
Type of Ecommerce



Business-to-Customer (B2C) – involves customers gathering information; purchasing goods or receiving products over electronic network

Examples:

- ✓ **Dells selling laptop to a user**



Type of Ecommerce



Business-to-Government (B2G) – it refers to use of internet for government procurement, licensing procedures, and other government related operations.

Examples:

- ✓ **Business paying taxes, file reports, or sell goods and services to Govt. agencies**



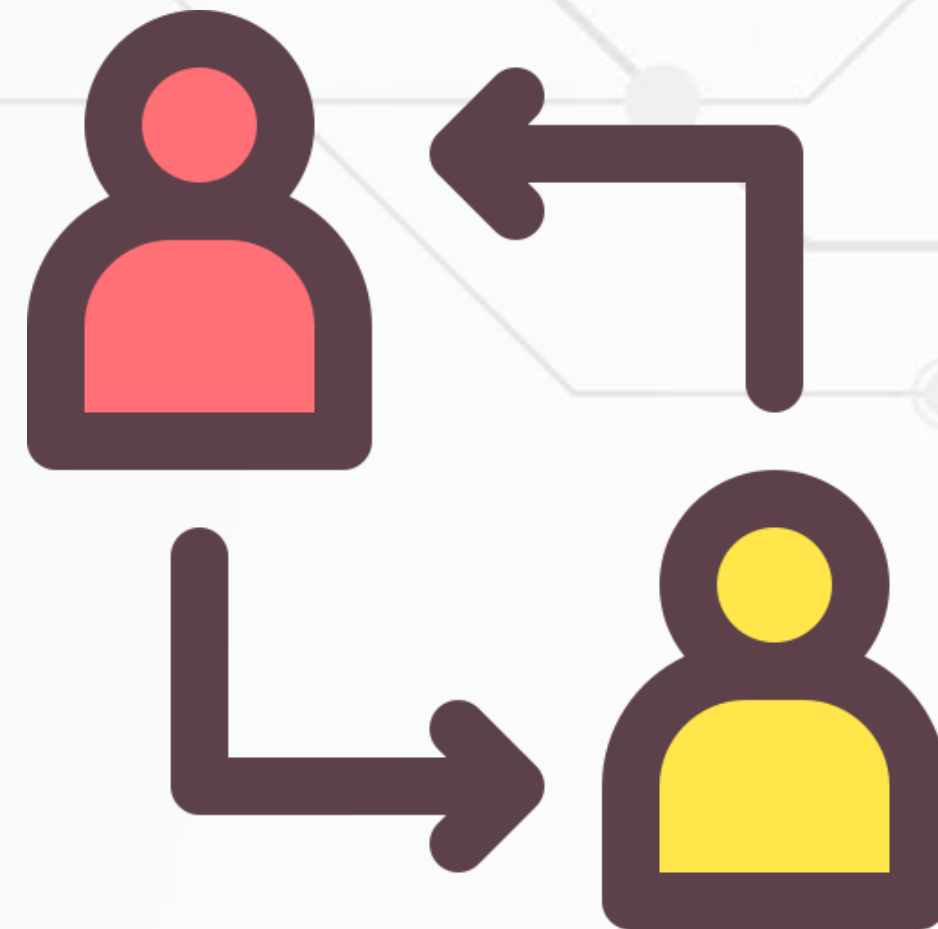
Type of Ecommerce



Customer-to-Customer (C2C) – is simply commerce between private individuals or customers

Examples:

- ✓ **JL buying an iPhone from Akira in Ebay**
- ✓ **Gelo selling a car to Mikki**



Type of Ecommerce



**Government-to-Citizen (G2C) –
government to provide good and effective
services to each citizens**

Examples:

- ✓ JL applies an NBI Clearance and pay through the NBI Clearance Website



Type of Ecommerce



Government-to-Business (G2B) – a business model that refers to government providing services to business organization

Examples:

- ✓ **Websites support auctions, tenders and application submission functionalities**

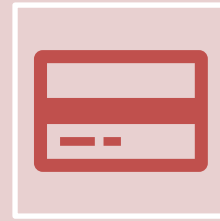


Application of Ecommerce

- ✓ **Online Shopping**
- ✓ **Online Banking**
- ✓ **Electronic Bills Payment**
- ✓ **Electronic Ticketing**



Common Threats in Ecommerce



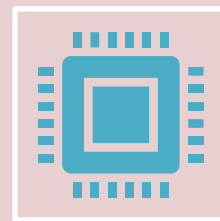
Financial Fraud: This is the most common threat, involving the use of stolen credit card information or fraudulent chargeback requests.



Malware: Malicious software like viruses, spyware, and ransomware can be used to disrupt operations, steal data, or take control of a system.



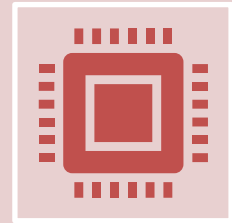
Phishing: Attackers trick customers or employees into revealing sensitive information, like login credentials or payment details, by impersonating a legitimate business.



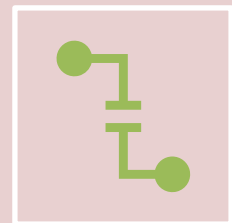
Brute-Force Attacks: These are automated attempts to guess passwords, often targeting admin panels or customer accounts with weak credentials.



Common Threats in Ecommerce



SQL Injection: Hackers exploit vulnerabilities in a website's database to access, modify, or delete sensitive data, including customer records.



DDoS Attacks (Distributed Denial of Service): These attacks overwhelm a website with a flood of traffic, making it inaccessible to legitimate users and causing significant downtime and lost sales.



E-skimming: This involves injecting malicious code onto a website's checkout page to steal credit card information as customers enter it.



Security in Ecommerce



SSL/TLS Certificates: These certificates encrypt the data transmitted between a customer's browser and the server, ensuring sensitive information like credit card numbers is protected. Look for "HTTPS" and a padlock icon in the browser's address bar.

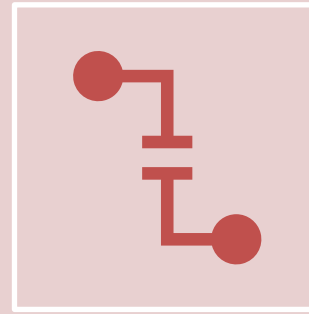


Secure Payment Gateways: Don't store customer credit card information directly on your servers. Instead, use a reputable third-party payment gateway (like Stripe or PayPal) that handles the transaction securely and is PCI DSS (Payment Card Industry Data Security Standard) compliant.

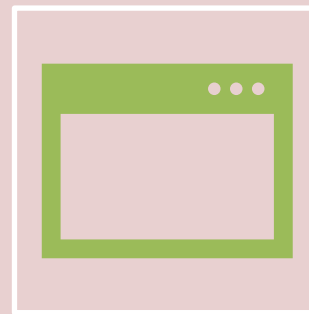


Strong Passwords and Multi-Factor Authentication (MFA): Enforce strong password policies for both customers and employees. MFA adds an extra layer of security by requiring a second form of verification (like a code sent to a phone) in addition to a password.

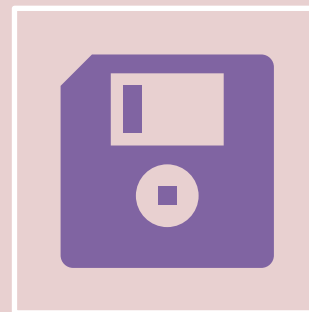
Security in Ecommerce



Firewalls: A firewall acts as a barrier, monitoring and filtering incoming and outgoing network traffic to block suspicious activity and attacks like SQL injections and DDoS attacks.



Regular Software Updates: Keep your e-commerce platform, plugins, and themes up to date. Updates often include security patches that fix known vulnerabilities that hackers could exploit.



Regular Backups: Regularly back up your entire website and its database. In the event of a security breach or data loss, a recent backup can be a lifesaver, allowing for a quick recovery with minimal data loss.